
Error Structure Determines Correctness Preference in Contradictory Training Data

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Abstract

1 Language models trained on contradictory data often prefer correct answers, yet
2 the mechanism behind this preference is poorly understood. Without such under-
3 standing, we cannot predict when this implicit filtering will fail—a question crit-
4 ical for controlling model behavior on noisy real-world corpora. We hypothesize
5 that next-token prediction, as a compression process, favors whichever answer
6 cluster has lower description length; truth benefits only when errors lack inter-
7 nal structure. We train transformers (3.5M–1B parameters) from scratch on con-
8 trolled corpora where the same problem appears with both correct and incorrect
9 solutions, systematically varying the structure of errors. We demonstrate three
10 findings: (a) when errors are random, models develop a correctness preference
11 scaling from 65% to 85% with model size; (b) when errors follow a single coher-
12 ent alternative rule, this preference vanishes entirely ($\sim 45\text{--}51\%$); (c) the transi-
13 tion is sharp—two competing wrong rules suffice to restore correctness preference
14 ($47\% \rightarrow 78\%$). The pattern reproduces on real Wikipedia text with entity substitu-
15 tion (71% vs 46%) and at 1B scale on real web text (77% vs 47%). These results
16 support the hypothesis that, in controlled contradictory corpora, model preference
17 is driven by the relative compressibility of competing answer systems rather than
18 by truth per se.

19 1 Introduction

20 Real-world training corpora are noisy: the same question may receive contradictory answers across
21 different documents. Yet language models trained on such data tend to prefer correct information—
22 and this implicit filtering is increasingly relied upon in high-stakes applications from medical QA to
23 legal reasoning. If we do not understand *why* this filtering works, we cannot predict *when* it will fail.
24 This is not merely a theoretical concern: coordinated misinformation campaigns, systematic biases
25 in training data, and internally consistent pseudoscientific frameworks all present cases where errors
26 may share enough structure to evade whatever mechanism produces truthful preferences.

27 Several explanations have been offered. Scaling improves factual performance [Kadavath et al.,
28 2022]. RLHF steers outputs toward human preferences. Frequency matters: factual accuracy cor-
29 relates with how often correct information appears in training data [Elazar et al., 2022, Joshi et al.,
30 2024, Kandpal et al., 2023]. Recent work has found truth-correlated internal representations [Burns
31 et al., 2023, Marks and Tegmark, 2024, Bürger et al., 2024] and established that calibrated models
32 must hallucinate at a rate bounded by the corpus’s monofact rate [Kalai and Vempala, 2024]. Yet
33 none of these explain why the training objective would favor one answer over another when both
34 appear at equal frequency in the same format.

35 We propose that the answer lies in the structure of errors. Minimizing cross-entropy is equivalent
36 to minimizing code length [Shannon, 1948, Deletang et al., 2024], linking LLM training to the

37 Minimum Description Length principle [Rissanen, 1978, Grünwald, 2007]. A correct rule system
 38 compresses into a compact representation; diverse errors must be memorized individually. But a *co-*
 39 *herent* false system—internally consistent, just wrong—compresses equally well, and the preference
 40 vanishes.

41 We test this hypothesis in a controlled experimental setting. We train transformers (3.5M–1B pa-
 42 rameters) from scratch on corpora where each mathematical problem appears with both correct and
 43 incorrect solutions, systematically varying the structure of errors. This design isolates error structure
 44 as the independent variable while holding frequency, format, and domain constant.

45 Our contribution is a single experimentally validated hypothesis: **in controlled contradictory cor-
 46 pora, the compression objective favors consistency, not truth.** We show that (a) random errors
 47 produce a correctness preference scaling from 65% to 85%; (b) a single coherent alternative rule
 48 eliminates this preference entirely; (c) two competing rules restore it, pinpointing the compressibil-
 49 ity boundary; (d) the same contrast reproduces on real Wikipedia text; and (e) the effect persists at
 50 1B parameters on real web text (Section 4.7). Whether this principle extends to large-scale pretrain-
 51 ing remains an open question that we discuss in Section 5.

52 2 Related Work

53 **Prediction as compression.** The link between prediction and compression traces to Shannon
 54 [1948] and was formalized by Solomonoff [1964] and Rissanen’s MDL principle [Rissanen, 1978,
 55 Grünwald, 2007]. Deletang et al. [2024] showed that LLMs are universal compressors. Huang et al.
 56 [2024] discovered a linear correlation ($r \approx -0.93$) between compression quality and benchmark per-
 57 formance. Wan and Mei [2025] proved that LLM training approximates Solomonoff induction. Pan
 58 et al. [2025] linked compression to knowledge acquisition and scaling; Chlon et al. [2025] linked
 59 compression failures to hallucination. We build on the MDL framework by experimentally varying
 60 the description length of false answer systems.

61 **Internal representations and world models.** Compression can give rise to structured internal
 62 representations: Li et al. [2023a] found board representations in an Othello model, Gurnee and
 63 Tegmark [2024] discovered linear space-time representations in Llama-2, and several studies found
 64 truth-correlated representations [Marks and Tegmark, 2024, Burns et al., 2023, Li et al., 2023b,
 65 Azaria and Mitchell, 2023, Ravfogel et al., 2025]. Our work operates at the behavioral level: we
 66 identify data-level conditions under which compression produces a preference for correct comple-
 67 tions, leaving activation-level analysis for future work.

68 **Truthfulness and data statistics.** Joshi et al. [2024] linked truthfulness to “personas” in pretrain-
 69 ing data. Elazar et al. [2022] demonstrated frequency dependence. Kandpal et al. [2023] showed a
 70 direct relationship between document count and accuracy. Li et al. [2024] studied LLM preferences
 71 under conflicting knowledge and found that formality and surface quality predict which version
 72 the model favors—a consistency-driven explanation complementary to ours. A growing body of
 73 work studies knowledge conflicts directly: Xie et al. [2024] surveyed approaches to resolving con-
 74 flicts between parametric and contextual knowledge, and Longpre et al. [2021] showed that entity-
 75 substituted evidence can override parametric memory. Unlike these analyses, we fix frequency,
 76 format, and surface quality, varying only the compressibility of the error system itself—isolating
 77 error structure as the causal variable rather than studying post-hoc conflict resolution.

78 **Simplicity bias and noisy labels.** Neural networks prefer simple functions [Valle-Perez et al.,
 79 2019, Mingard et al., 2021, Goldblum et al., 2024]. The noisy labels literature directly paral-
 80 lels our setup: Zhang et al. [2017] showed memorization of random labels; Rolnick et al. [2017]
 81 showed robustness to massive label noise. Grokking connects to compression as a memorization-to-
 82 generalization transition [Nanda et al., 2023, Liu et al., 2023]. Our experiments extend this line by
 83 showing that “structured noise”—coherent errors—is not filtered out. To our knowledge, systematic
 84 variation of error compressibility in a denoising setting has not been studied before.

85 3 Methodology

86 **Overview.** We test the hypothesis that next-token prediction favors whichever answer system has
 87 lower description length, regardless of its truth value. To isolate this, we train models from scratch

on controlled corpora where each problem appears with contradictory answers, varying only the structure of errors. The core experiments compare random errors (high description length) against coherent errors (low description length) at equal frequency (Section 4.1), then probe the boundary with multi-rule errors of intermediate compressibility (Section 4.2). Transfer to natural language is tested via Wikipedia entity substitution (Section 4.3).

3.1 Hypothesis and MDL Framing

Consider a corpus where each problem appears with a correct answer (theory T_1) and an alternative (T_2). An MDL learner [Rissanen, 1978, Grünwald, 2007] minimizes $L(M) + L(D|M)$. When $K(T_2) \gg K(T_1)$ (random errors), the false system’s description length grows with corpus size, favoring T_1 . When $K(T_2) \approx K(T_1)$ (coherent errors), both are compact and the learner has no basis to prefer one over the other. With N competing false rules, a random “selector” (which rule applies to which problem) adds $\log N$ bits per problem, progressively degrading the false cluster’s compressibility.

Throughout, “truth” means correctness of mathematical derivations and factual accuracy of Wikipedia paragraphs—models compress text, not reality. The compressibility gap is domain-dependent and smaller in natural language than in formal math (Section 4.3). Additional limitations are discussed in Section 6.

3.2 Models and Training

Custom decoder-only transformers with GPT-2-like architecture (pre-norm, GELU activation, causal mask). Size labels below are internal designations and do not correspond to OpenAI’s GPT-2 checkpoints (e.g., our “large” at 86M is smaller than GPT-2 Small at 117M):

Config	Layers	d_{model}	Heads	Parameters
tiny	4	256	4	~3.5M
small	6	384	6	~12M
medium	8	512	8	~26M
large	12	768	12	~86M

Table 1: Model configurations used in all experiments.

Optimizer: AdamW (weight_decay=0.01), cosine decay with linear warmup, lr = $3\text{e-}4$, seq_len = 256, batch_size = 32, 5000 steps. All experiments use 4 random seeds for core conditions (2 where noted). Learning curves confirm behavioral plateau by step 3000–4000 at all sizes (Appendix F). Denoising experiments use PyTorch; multi-rule and Wikipedia experiments use MLX. All problems are generated and verified by SymPy; no human annotation is involved. Full generation templates are provided in the supplementary material.

3.3 Corpus Design

Denoising setup (primary). A generator creates mathematical problems of four types: multi-step arithmetic, factoring, equation solving, and differentiation, formatted as step-by-step derivations in English. The key design: **each problem appears with contradictory answers**—modeling the scenario where the same question receives conflicting responses. The tokenizer is character-level (vocab ~57); BPE robustness is verified in Section 4.4.

Condition	Correct	Incorrect	Ratio	Purpose
Equal random	1	1 random	1:1	Signal extraction from random noise
Equal coherent	1	1 coherent	1:1	Control: coherent errors eliminate bias
2:1 noise	1	2 random	1:2	Noise tolerance at moderate noise
4:1 noise	1	4 random	1:4	Noise tolerance at high noise

Table 2: Denoising corpus conditions. Each corpus contains 5,000 unique problems.

Size	Params	Random Accuracy	Coherent Accuracy	Random Seeds	Coherent Seeds
tiny	3.5M	65.3% $\pm$ 1.3%	43.5% $\pm$ 2.6%	4	4
small	12M	74.6% $\pm$ 1.6%	44.5% $\pm$ 3.0%	4	4
medium	26M	81.1% $\pm$ 1.2%	45.8% $\pm$ 3.4%	4	4
large	86M	85.2% $\pm$ 2.3%	51.0% $\pm$ 0.8%	2	2

Table 3: Denoising paired evaluation: equal random vs equal coherent errors, across model sizes. Random accuracy scales with model size; coherent accuracy remains near chance.

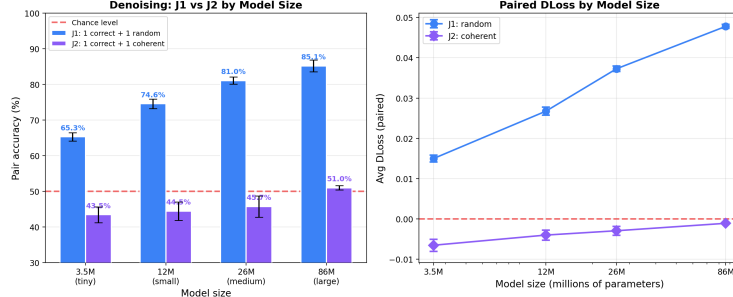


Figure 1: The central contrast. Random errors: accuracy scales from 65% to 85%. Coherent errors: accuracy stays near chance across all sizes.

Wikipedia setup (transfer). To test generalization beyond formal math, we construct corpora from 20,000 Wikipedia paragraphs. Using spaCy NER, we create two corruption modes: *random substitution* (each entity replaced with a random entity of the same type) and *coherent substitution* (a consistent global mapping, e.g., every “France” $\rightarrow$ “Japan”, every “Paris” $\rightarrow$ “Tokyo”). Coherent substitution preserves grammatical structure and cross-entity consistency within each paragraph; random substitution breaks co-occurrence patterns (e.g., a paragraph may mention “Kumamoto” in a context about French history). Both modes substitute entities of matching type, so surface-level grammatical anomalies (gender, case agreement) are minimal. Models train on 50/50 mixes; evaluation uses 2,000 held-out paired paragraphs.

3.4 Evaluation

Paired evaluation (primary). For each problem, a shared prompt is generated with two completions (correct and incorrect). NLL is computed on completion tokens only, conditioned on the prompt. The primary metric is **pair accuracy**: the fraction of pairs where the model assigns lower NLL to the correct completion. This is equivalent to the Common Language Effect Size (CLES; McGraw and Wong, 1992). Length-matched evaluation confirms that accuracy is not driven by completion length differences (within 1 pp across all conditions).

Generative evaluation. Greedy decoding on 500 test prompts with automated SymPy verification. This confirms that the discriminative effect extends to generation (Section 4.5; Table 6).

4 Results

4.1 The Central Contrast: Random vs Coherent Errors

When the same problem appears with both a correct and a random wrong answer, the model learns to prefer the correct one—accuracy reaches 85% at 86M parameters (Table 3, Figure 1). When the wrong answer follows a coherent rule system, the effect disappears: accuracy hovers near 50% at all scales.

Random accuracy increases monotonically with scale (65% $\rightarrow$ 75% $\rightarrow$ 81% $\rightarrow$ 85%). Coherent accuracy converges toward 50% from below, consistent with the MDL prediction that equally compressible systems at equal frequency should be indistinguishable. The below-chance coherent accuracy at tiny ($\sim$ 43%) reflects textual simplicity of the false rules (e.g., dropping terms in deriva-

N Rules	Accuracy	p
1 (coherent)	46.6% $\pm$ 2.4%	~ 1.0
2	77.6% $\pm$ 1.3%	$< 10^{-6}$
3	82.8% $\pm$ 0.1%	$< 10^{-6}$
5	84.8% $\pm$ 0.5%	$< 10^{-6}$
10	88.3% $\pm$ 0.7%	$< 10^{-6}$

Table 4: Multi-rule paired evaluation (tiny, 3.5M, 50/50, 4 seeds). $N=2$ on small (12M) yields 86.3% $\pm$ 0.8%, confirming the effect strengthens with capacity.

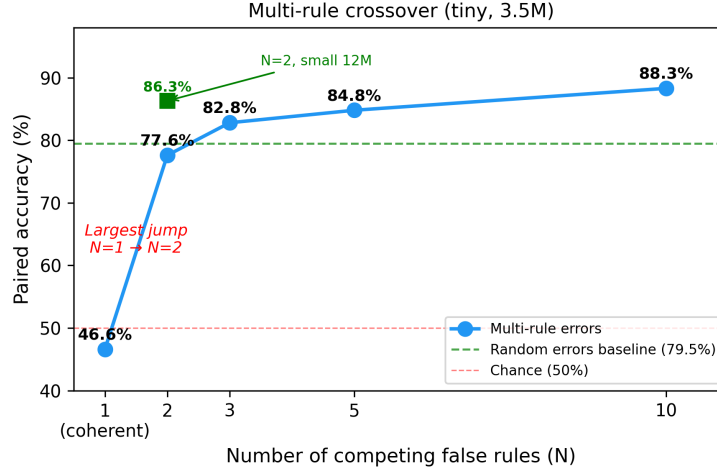


Figure 2: Multi-rule crossover. Accuracy jumps from 47% ($N=1$) to 78% ($N=2$), then grows gradually through $N=10$.

149 tives); the compressor prefers whichever system is simpler to encode. The multi-rule experiment
150 (Section 4.2) controls for this surface-form effect.

151 4.2 Multi-Rule Errors: The Sharp Crossover

152 The denoising experiments establish two poles: one coherent rule yields chance, random errors
153 yield strong bias. We probe the boundary by training on corpora with N alternative wrong rules per
154 task type, where each problem’s rule is chosen at random. Each rule is compact, but the mapping
155 “problem $\rightarrow$ rule” is unpredictable.

156 This is the most mechanistically revealing result (Table 4, Figure 2). With one false rule, $K(T_2) \approx$
157 $K(T_1)$. With two rules, the learner must encode a random selector—which rule applies to which
158 problem—adding high-complexity bits that break compressibility. At $N=10$, accuracy (88.3%)
159 exceeds even the random baseline (79.5%), because multi-rule errors combine compactness of indi-
160 vidual rules with incompressibility of the selector.

161 4.3 Transfer to Real Text: Wikipedia

162 The random/coherent contrast reproduces (Table 5): random substitution yields 70–71% accuracy
163 (all $p < 10^{-100}$), coherent substitution yields 46–49% (at or below chance). The effect is weaker
164 than in math (71% vs 85%) because natural language provides more textual flexibility for locally
165 fluent errors. Unlike math, Wikipedia accuracy saturates near 70% across the tested range—the
166 compressibility gap in natural language does not substantially benefit from additional capacity at
167 this scale. Per-entity-type accuracy ranges from 82% (LOC) to 61% (CARDINAL), with geographic
168 entities showing the strongest effect (Appendix E).

Size	Random Accuracy	Coherent Accuracy
tiny (3.5M)	69.6% $\pm$ 0.1%	48.7% $\pm$ 0.5%
small (12M)	70.7% $\pm$ 0.4%	46.6% $\pm$ 0.4%
medium (26M)	71.5% $\pm$ 0.8%	46.4% $\pm$ 0.8%
large (86M)	71.4% $\pm$ 0.7%	45.9% $\pm$ 1.5%

Table 5: Wikipedia entity substitution (50/50, 4 seeds per size). The random/coherent contrast reproduces on real text.

Size	Random-trained	Coherent-trained	Gap
tiny (3.5M)	30.5% $\pm$ 1.7%	20.8% $\pm$ 3.6%	+9.7 pp
small (12M)	46.7% $\pm$ 1.9%	31.7% $\pm$ 1.8%	+15.0 pp
medium (26M)	50.5% $\pm$ 1.3%	36.1% $\pm$ 2.7%	+14.4 pp
large (86M)	52.8% $\pm$ 1.1%	35.6% $\pm$ 1.8%	+17.2 pp

Table 6: Generative accuracy (4 seeds, 500 problems each). The gap widens with model size.

169 4.4 Robustness Checks

170 **Tokenization.** The primary experiments use a character-level tokenizer (vocab $\sim$ 57). To verify
171 that the effect is not an artifact of tokenization, we repeat the core conditions with a BPE tokenizer
172 (SentencePiece, vocab=1000). The effect survives and strengthens: random accuracy increases from
173 65.3% to 75.9% in denoising (79.5% to 85.6% in standard); coherent accuracy remains at chance
174 (49.3% and 45.9% respectively). Full results in Appendix F.

175 **Standard (non-denoising) setup.** In the standard setup, each problem appears once with either
176 a correct or incorrect solution (no within-problem contradiction). Paired accuracy is higher (79.5–
177 88.3% vs 65.3–85.2%), but the gap closes with scale (from 14 pp at tiny to 3 pp at large). Critically,
178 the random/coherent contrast holds identically: standard coherent accuracy is 47–52%, at chance.
179 Full comparison in Appendix D.

180 **Frequency vs compressibility.** For random errors, truth bias persists even when correct examples
181 are a small minority: 67% paired accuracy at 10/90 correct-to-incorrect ratio (Appendix D). For
182 coherent errors, the pattern reverses—frequency alone determines preference: at 40/60 coherent, the
183 model follows the majority with 72% preference for the false system; at 20/80, this rises to 91%.
184 This asymmetry is a direct prediction of the MDL framework: when both systems compress equally
185 well, frequency is the only tiebreaker.

186 **Matched-control ablation.** A potential confound is that coherent errors affect more derivation
187 steps per problem ($\sim$ 3 on average) than standard random errors (exactly 1). To control for this, we
188 generate “matched random” errors that corrupt the same number of steps as coherent errors but at
189 random positions with random values. Matched-random accuracy is 74.5% $\pm$ 0.4% (4 seeds)—lower
190 than standard random (79.5%, which corrupts only 1 step) but far above coherent (47.2%, which also
191 corrupts $\sim$ 3 steps). The 27 pp gap between matched-random and coherent at identical step counts
192 confirms that compressibility, not surface-level corruption load, drives the random/coherent contrast.

193 **Noise tolerance.** The correct signal persists under increasing noise ratios (1:2 and 1:4 random),
194 with accuracy degrading gracefully from 85% to 66–75% at large scale. Details in Appendix D.

195 4.5 Generative Evaluation

196 Paired evaluation is a forced-choice (discriminative) setting. To verify that the effect extends to
197 generation, we run greedy decoding with SymPy verification on 500 problems per model across all
198 sizes.

199 The gap widens with model size (10 pp at tiny $\rightarrow$ 17 pp at large), confirming that the discriminative
200 preference translates into a generative advantage (Table 6). The absolute generation accuracy is
201 substantially lower than paired accuracy (53% vs 85% at large). This gap reflects the fundamental
202 difference between discrimination and generation: the model may assign higher likelihood to correct

Condition	seed43	seed44	seed45	Mean
Random	85.3%	85.7%	89.3%	86.8%
Coherent	49.3%	52.8%	49.8%	50.6%

Table 7: Qwen3-0.6B paired evaluation (50/50, 3 seeds). The random/coherent contrast reproduces across architectures.

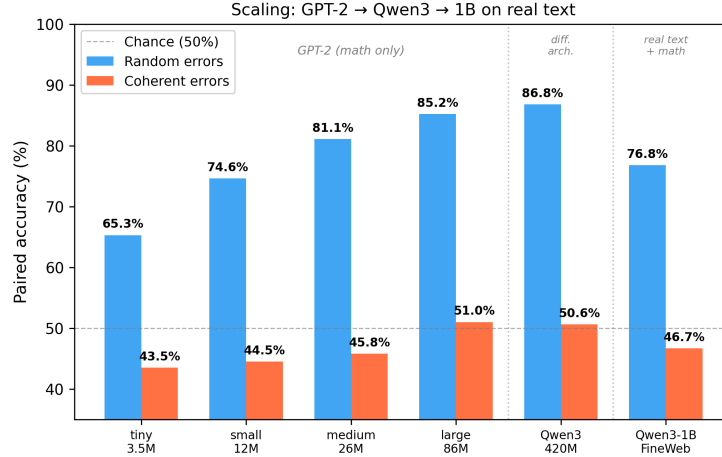


Figure 3: Full scaling results. GPT-2 family (3.5M–86M, math-only), Qwen3-0.6B (420M, different architecture, math-only), and a Qwen3-architecture model (~1B, FineWeb-Edu + 8% math).

203 completions while lacking the capacity to produce them reliably from scratch. Both metrics agree
204 on the direction: random-trained models outperform coherent-trained ones at all scales.

205 Per-task breakdown reveals that generation difficulty varies sharply by type. For random-trained
206 large models: derivative (79.5%), algebra (70.4%), equation (61.0%), arithmetic (0%). Multi-step
207 arithmetic requires exact carry propagation across many digits—a well-known difficulty for au-
208 toregressive generation at this scale—even though paired evaluation shows strong discrimination.
209 Coherent-trained models show lower generation accuracy across all types, with the gap largest for
210 algebra (70.4% vs 38.3%) and smallest for derivative (79.5% vs 56.5%).

211 4.6 Architecture Robustness: Qwen3

212 To verify that the effect is not specific to the GPT-2 architecture, we train a Qwen3-0.6B model
213 (420M parameters, 28 layers, RoPE + GQA + SwiGLU + RMSNorm) from scratch on the same
214 denoising corpora. This architecture differs from GPT-2 in positional encoding, attention mecha-
215 nism, activation function, and normalization—testing whether the compression-consistency effect is
216 a general property of autoregressive transformers.

217 The pattern reproduces (Table 7, Figure 3): random errors yield 86.8% accuracy, coherent errors
218 yield 50.6% (chance). The random/coherent contrast is architecture-independent. Qwen3’s random
219 accuracy (86.8%) is comparable to GPT-2 large (85.2%) despite training on the same corpus sized
220 for smaller models, suggesting further gains with compute-matched training.

221 4.7 Scaling to 1B on Real Text

222 The preceding experiments use synthetic math corpora and models up to 86M parameters (420M for
223 architecture robustness). A natural objection is that the effect may be specific to toy-scale models or
224 synthetic data. To address both concerns simultaneously, we train a Qwen3-architecture model (~1B
225 parameters) from scratch on 1 GB of FineWeb-Edu [Penedo et al., 2024]—a curated subset of real
226 web text—with mathematical content filtered out and replaced by 8% contradictory math problems
227 (denoising format, equal random or coherent). This setup tests whether the compression-consistency
228 effect survives in a regime where (a) the model has two orders of magnitude more parameters than

Condition	Accuracy	ΔLoss	p	Seeds
Random	76.8%	+0.098	$< 10^{-6}$	1
Coherent	46.7%	-0.003	~ 1.0	1

Table 8: Qwen3-architecture $\sim 1\text{B}$ trained on FineWeb-Edu + 8% contradictory math (denoising, 50/50). The random/coherent contrast reproduces at 1B scale on real text.

our primary experiments, and (b) the contradictory signal is a small fraction of a large, naturalistic corpus.

The result (Table 8) confirms that the random/coherent contrast holds: random errors yield 76.8% paired accuracy ($\Delta\text{Loss} = +0.098$, $p < 10^{-6}$), while coherent errors yield 46.7% (chance). The random accuracy is lower than the math-only GPT-2 large (85.2%) and Qwen3-0.6B (86.8%), likely because the contradictory math signal constitutes only 8% of the training corpus, diluted by general text. The key finding is that even in this diluted, naturalistic setting, the compression objective still distinguishes random from coherent errors—and the ΔLoss magnitude (+0.098) is the largest we observe, suggesting that the 1B model develops a stronger per-pair preference despite lower aggregate accuracy.

5 Discussion

Summary of findings. The experiments support a unified hypothesis: **in our controlled settings, the compression objective tracks consistency rather than truth.** Any internally consistent rule system—true or false—compresses equally well. Truth bias emerges only when false alternatives are structurally incoherent. The evidence converges across setups (denoising, standard, Wikipedia), error types (random, coherent, multi-rule), and evaluation modes (discriminative and generative).

The role of frequency. Our results reveal an asymmetry between random and coherent error regimes. For random errors, compressibility overrides frequency: truth bias persists even at 10/90 correct-to-incorrect ratio (67% accuracy). For coherent errors, frequency is decisive: at 40/60 coherent mixing, the model follows the majority with 72% preference for the false system, rising to 91% at 20/80. This is a direct MDL prediction—when competing systems compress equally well, the more frequent one wins. The practical implication is that coherent falsehood needs no frequency advantage to neutralize truth bias; equal frequency suffices.

What the multi-rule experiment shows. The sharp $N=1 \rightarrow 2$ transition is the strongest evidence that compressibility, not surface-form complexity, is the operative variable. Each individual rule at $N=2$ remains compact, yet the random assignment of rules to problems introduces incompressible bits. This rules out explanations based on lexical diversity or number of corrupted tokens alone.

Implications and scope. In our controlled settings, the training objective does not provide an inherent “truth compass.” Internally coherent falsehood remains competitive. Embedding cross-domain verification can partially restore truth bias, but the effect exhibits inverse scaling: verification accuracy drops from 71% at 3.5M to 61% at 86M parameters (Appendix F), as larger models absorb coherent within-domain patterns more readily. This inverse scaling is verified across 10 models (4 tiny + 4 small + 2 large) with consistent results. This has potential relevance for alignment (systematic false beliefs may resist filtering and grow harder to verify at scale), hallucinations (coherent misconceptions may persist; cf. Kalai and Vempala, 2024), and data curation (diverse errors are filtered effectively, but coordinated errors may not be). The 1B experiment on real web text (Section 4.7) provides initial evidence that the effect persists beyond toy scale and synthetic data, but all our experiments remain far below the scale of production LLMs. Whether these findings extend to large-scale pretraining on heterogeneous real-world corpora remains an open question. We view our results as identifying a mechanism and its boundary conditions, not as direct claims about production LLMs.

Our findings admit an interpretive analogy with Popper’s falsifiability criterion [Popper, 1959]: a false theory that requires ad hoc corrections to accommodate observations has higher description length than a true one. The analogy is limited—the model does not “test” theories—but the structural parallel is suggestive.

6 Conclusion

Whether the compression mechanism that produces truthful preferences in language models extends to large-scale pretraining remains an open question. In our controlled experiments with transformers up to 86M parameters—and up to ~1B parameters on real web text (Section 4.7)—we find that models prefer the correct answer only when errors are structurally incoherent; a single coherent alternative rule eliminates this preference, and two competing rules restore it. The same pattern reproduces on real Wikipedia text, in generative evaluation, across architectures, and at 1B scale on naturalistic corpora. In the settings we study, correctness preference is a compression artifact whose presence or absence is determined by the description length of the false answer system.

Limitations

Model scale. The primary experiments use models of 3.5M–86M parameters. Section 4.7 extends to a Qwen3-architecture model (~1B parameters) trained on real text, confirming the random/coherent contrast at larger scale. However, this remains far below the scale of production LLMs (7B–400B+), and the 1B result uses only a single seed per condition. For coherent errors, an MDL argument suggests the result should hold regardless of scale: equally compressible systems at equal frequency offer no basis for preference. Replication at multi-billion scale with compute-matched training remains an important next step.

Domain specificity. Mathematics provides an unusually crisp correct/incorrect distinction. The effect weakens in natural language (71% vs 85%), where errors can remain locally fluent. Extending to domains with competing real-world knowledge systems (e.g., conflicting news sources, historical revisionism) remains open.

Discriminative vs generative gap. Paired accuracy (85%) substantially exceeds generative accuracy (53%) at large scale. This gap matters: real-world LLM behavior depends on generation, not discrimination. Several factors may contribute: generation requires producing correct tokens autoregressively (errors compound), while paired evaluation only requires relative likelihood ranking. The gap narrows with model size (49 pp at tiny, 32 pp at large), suggesting that scale partially bridges it. Critically, both metrics agree on the direction and on the random/coherent contrast at all tested scales—the discriminative preference is not an evaluation artifact. Still, paired accuracy should be interpreted as a controlled diagnostic of model preference, not as a direct measure of output truthfulness.

Seed counts. Core conditions use 4 seeds; some conditions use 2. Sufficient for directional stability, not for tight confidence intervals.

Causal identification. While we vary error structure as the independent variable, random and coherent corruptions may differ in additional ways (number of affected derivation steps, surface-form simplicity). The multi-rule experiment partially addresses this by holding individual rule complexity constant, but a fully matched control (identical corruption load, lexical diversity, and output length) would strengthen the causal claim.

Future directions. Linear probing for “truth directions” vs “coherence directions” in models trained under our conditions could connect behavioral findings to the internal representation literature. Testing interactions with RLHF and investigating whether the compressibility mechanism interacts with in-context learning are natural extensions.

Ethical Considerations

This work demonstrates that, in controlled settings, internally consistent misinformation may be harder for language models to filter than diverse errors. The finding does not directly demonstrate vulnerability in production models. We believe transparent reporting of conditions under which truth bias fails is more beneficial than concealment, as it can inform defensive measures in data curation and model evaluation.

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388 A Reproducibility

389 All code, data generation scripts, and evaluation scripts are available at [anonymous repository, in-
390 cluded as supplementary material]. Denoising experiments were run on Modal.com T4 GPUs using
391 PyTorch. Standard math and Wikipedia experiments were run on Apple Mac M4 (36GB) using
392 MLX (v0.31.0). Qwen3-0.6B experiments were run on Modal.com A10G GPUs; the ~ 1 B Qwen3-
393 architecture experiments were run on Modal.com A100 GPUs. Over 210 models were trained across
394 all conditions; total compute approximately 150 GPU-hours.

395 **Corpus generation examples.** Each problem is generated programmatically and verified by
396 SymPy. Below is a representative training example (denoising, equal random condition):

397 Problem: Expand and simplify $3 * (2x + 5) - 4x$
398 Step 1: $3 * (2x + 5) = 6x + 15$
399 Step 2: $6x + 15 - 4x = 2x + 15$
400 Answer: $2x + 15$

401 The corresponding random-error version replaces a derivation step with a plausible but incorrect
402 computation (e.g., $6x + 15 - 4x = 2x + 11$). In the coherent condition, a systematic rule is
403 applied consistently across all problems of the same type (e.g., for distribution: $a(b + c) = ab + c$
404 instead of $ab + ac$). The paired test set presents the same prompt with both correct and incorrect
405 completions for NLL comparison.

B Formal MDL Statement

We state the theoretical prediction that motivates our experiments more precisely.

Proposition (informal MDL prediction). Consider a corpus D containing n problems, each appearing with two answers: one generated by theory T_1 (correct) and one by theory T_2 (alternative). An ideal MDL learner selects the theory T^* minimizing total description length $L(T) + L(D|T)$.

- *Case 1 (random errors):* T_1 is a compact rule system with $L(T_1) = O(1)$. T_2 assigns independent random answers, requiring $L(T_2|D) = O(n)$. The MDL learner prefers T_1 for sufficiently large n .
- *Case 2 (coherent errors):* Both T_1 and T_2 are compact rule systems with $L(T_1) \approx L(T_2) = O(1)$. At equal frequency, $L(D|T_1) \approx L(D|T_2)$, and the MDL learner has no basis to prefer one over the other.
- *Case 3 (multi-rule, N competing false rules):* Each rule r_i is compact ($L(r_i) = O(1)$), but encoding the assignment of rules to problems requires an additional selector $s : \{1, \dots, n\} \rightarrow \{1, \dots, N\}$ with $L(s) = O(n \log N)$ when assignments are random. Total false-system description length grows as $O(n \log N)$, restoring the MDL advantage of T_1 .

This proposition is not a formal theorem about neural network training dynamics—it describes the behavior of an idealized MDL learner. Our experiments test whether gradient-descent-trained transformers approximate this behavior. The results in Sections 4.1–4.3 are consistent with all three predictions.

C Compression Measure (gzip)

To operationalize the MDL argument, we measure compression ratios (gzip level 9) on concatenated correct vs incorrect completions from each paired test set.

Condition	Correct	Incorrect	Delta	Paired Accuracy
random 50/50	0.1627	0.1639	+0.0012	79.5%
coherent 50/50	0.1656	0.1658	+0.0002	47.2%
contradictory	0.1745	0.1744	−0.0001	49.0%
multirule $N=2$	0.1671	0.1710	+0.0038	77.6%
multirule $N=5$	0.1672	0.1730	+0.0057	84.8%
multirule $N=10$	0.1676	0.1752	+0.0076	88.3%

Table 9: Compression ratio and paired accuracy across conditions. Spearman $\rho = 0.68$, $p = 0.042$.

Conditions with larger compression gaps produce stronger truth bias. We present this as supporting evidence; the primary argument rests on the experimental contrasts in Section 4.

D Standard (Non-Denoising) Experiments

D.1 Standard Corpus Design

In the standard setup, each problem appears once with either a correct or incorrect solution. The two groups do not share prompts.

D.2 Standard vs Denoising

The denoising setup is harder (within-problem contradiction vs across-corpus mixing), but the gap closes with scale.

Size	Standard	Denoising	Gap
tiny	79.5%	65.3%	−14.2 pp
small	86.2%	74.6%	−11.6 pp
medium	87.1%	81.1%	−6.0 pp
large	88.3%	85.2%	−3.1 pp

Table 10: Standard vs denoising accuracy (50/50 random). The gap closes with scale.

Proportion	Random	Coherent
50/50	80%	47.2%
40/60	79%	27.8%
30/70	75%	14.7%
20/80	69%	9.6%
10/90	67%	—

Table 11: Random vs coherent accuracy across proportions (standard, tiny). Random: truth bias persists at 10/90. Coherent: the model follows pure frequency.

437 D.3 Frequency Effects

438 D.4 Noise Tolerance

439 D.5 NLL Distribution

440 Random errors produce a right-skewed per-pair NLL difference (mean +0.048, median +0.025,
441 81.5% positive). Coherent errors produce a symmetric distribution (45.5% positive). This explains
442 why pair accuracy and mean Δ Loss can diverge.

443 E Additional Transfer Experiments

444 E.1 Synthetic World

445 A synthetic world with 50 entities, 4 types, 15 deterministic rules. Random 50/50: $57.7\% \pm 1.7\%$.
446 Coherent 50/50: $46.6\% \pm 1.7\%$. Per-type: minerals best (68.7%), potions near chance (49.1%).

447 E.2 Wikipedia Per-Entity-Type

448 E.3 Cross-Domain Falsification

449 Adding correct cross-domain tasks to a coherent corpus selectively increases derivative accuracy
450 ($35\% \rightarrow 56\%$ at 25%). The effect is non-monotonic (drops at 50% due to dilution).

451 F Robustness Checks

452 F.1 BPE Tokenization

453 F.2 Chained Verification

454 Embedding verification dependencies in coherent-error tasks restores truth bias: 70.9% at tiny (vs
455 43% standard coherent). The effect *decreases* with model size (64% at small, 61% at large)—larger
456 models absorb coherent errors more readily, making verification less effective.

457 F.3 Learning Curves

458 All sizes reach behavioral plateau by step 3000–4000. The large model achieves 88.8% at step 3000,
459 stable through 5000.

Condition	Ratio	Tiny	Small	Medium	Large
Equal random	1:1	65.3%	74.6%	81.1%	85.2%
2:1 noise	1:2	59.0%	68.6%	73.6%	75.2%
4:1 noise	1:4	56.6%	64.9%	66.6%	65.8%

Table 12: Paired accuracy at increasing noise ratios (denoising). Accuracy degrades gracefully; at 4:1 noise, it plateaus at medium/large.

Entity Type	<i>N</i> Pairs	Accuracy
LOC	45	82.2%
NORP	197	78.7%
GPE	301	77.1%
PERSON	402	69.9%
ORG	664	65.2%
CARDINAL	172	61.0%

Table 13: Wikipedia accuracy by entity type (random substitution, averaged across sizes).

G Per-Seed Details

NeurIPS Paper Checklist

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?
Yes. All claims are qualified to our controlled experimental setting.
2. **Limitations.** Does the paper discuss the limitations of the work performed by the authors?
Yes. See the Limitations section.
3. **Theory Assumptions and Proofs.** For each theoretical result, does the paper provide the full set of assumptions and a complete (or correct) proof?
Yes. The MDL prediction (Appendix B) is stated as an informal proposition with explicit assumptions. It describes an idealized MDL learner, not a formal theorem about neural networks.
4. **Experimental Result Reproducibility.** Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?
Yes. All hyperparameters, model architectures, data generation procedures, and evaluation protocols are specified. Code is provided as supplementary material.
5. **Open access to data and code.** Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?
Yes. Anonymous repository included as supplementary material.
6. **Experimental Setting/Details.** Does the paper specify all the training and test details necessary to understand the results?
Yes. See Section 3 and Appendix A.
7. **Experiment Statistical Significance.** Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
Yes. Standard deviations across seeds and *p*-values (binomial test) are reported for all conditions.
8. **Experiments Compute Resources.** For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?
Yes. See Appendix A: approximately 150 GPU-hours total on T4/A10G/A100 GPUs and M4 Apple Silicon.
9. **Code Of Ethics.** Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?
Yes.
10. **Broader Impacts.** Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
Yes. See Ethical Considerations section.

Setup	Tokenizer	Random	Coherent
Standard	Char	79.5%	47.2%
Standard	BPE	85.6%	45.9%
Denoising	Char	65.3%	43.5%
Denoising	BPE	75.9%	49.3%

Table 14: BPE vs char-level (tiny, 4 seeds). The effect survives BPE and strengthens for random errors.

Size	Seed 1	Seed 2	Seed 3	Seed 4	Mean
tiny	64.2%	64.1%	67.0%	65.8%	65.3%
small	75.1%	76.3%	72.6%	74.2%	74.6%
medium	79.6%	80.9%	82.4%	81.3%	81.1%
large	83.5%	86.8%	—	—	85.2%

Table 15: Denoising equal random, per-seed accuracy.

- 495 11. **Safeguards.** Does the paper describe safeguards that have been put in place for responsible
496 release of data or models that have a high risk for misuse?
497 **N/A.** The models are small ($\leq 1\text{B}$) and trained on synthetic or filtered web data; they pose no
498 misuse risk.
- 499 12. **Licenses for existing assets.** Are the creators or original owners of assets used in the paper prop-
500 erly credited and are the license and terms of use explicitly mentioned and properly respected?
501 **Yes.** Wikipedia data is used under its CC BY-SA license; all software dependencies are open
502 source.
- 503 13. **New Assets.** Are new assets introduced in the paper well documented and is the documentation
504 provided alongside the assets?
505 **Yes.** Code and data generation scripts are documented in the supplementary material.
- 506 14. **Crowdsourcing and Research with Human Subjects.** For crowdsourcing experiments and
507 research with human subjects, does the paper include the full text of instructions given to partici-
508 pants and screenshots?
509 **N/A.** No human subjects involved.
- 510 15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Sub-**
511 **jects.** Does the paper describe potential risks incurred by study participants?
512 **N/A.** No human subjects involved.

Size	Seed 1	Seed 2	Seed 3	Seed 4	Mean
tiny	40.2%	44.3%	46.3%	43.0%	43.5%
small	44.8%	45.3%	40.3%	47.4%	44.5%
medium	45.8%	40.9%	47.7%	48.6%	45.8%
large	51.6%	50.4%	—	—	51.0%

Table 16: Denoising equal coherent, per-seed accuracy.